Ecotopia: Fine-Tuning Mistral for Political Simulation

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1. Project Overview

Ecotopia is a political simulation game. The player is mayor of a city facing ecological collapse: 7 rounds (5 years each), a tile-based city grid, and three resource bars (ecology, economy, research). Each round, the player gives a speech with promises that directly affect citizens and resources.

We fine-tuned **small Mistral models** to handle two structured JSON tasks in the game pipeline:

- **FT-Extract**: Takes the player's raw speech text and outputs structured JSON with every promise identified, its category (ecology/economy/research), whether it helps or hurts that

resource, a deadline in game rounds, and any contradictions between promises. This feeds directly into the game engine to update resource bars.

- **FT-Citizens**: Takes the extracted promises plus the current game state (resource levels, citizen trust scores, round number) and generates dynamic citizen reactions. Each citizen has a name, archetype (environmentalist, industrialist, scientist...), a mood, spoken dialogue, and a trust delta that shifts their relationship with the player.

We test whether small fine-tuned models can match or beat **Mistral Large** on structured output, at a fraction of the cost.

2. Technical Approach

QLoRA + SFT

- **LoRA**: Trainable matrices (rank=16, alpha=32) injected alongside frozen weights. Only ~0.5% of parameters trained.
- **QLoRA**: Base model quantized to 4-bit NF4, then LoRA applied. Fits a 24B model on a single 48GB GPU.
- **SFT**: 690 input/output pairs generated via AWS Bedrock (Mistral Large). Under 10 minutes training per model.

Training Data

- Extraction: 300 examples (270 train / 30 val), three difficulty levels
- Citizens: 390 examples (351 train / 39 val), diverse scenarios and citizen types
- Hosted on HuggingFace: mistral-hackaton-2026/ecotopia-extraction-data and ecotopia-citizens-data

3. Models

FT-Extract

- **Ministral 8B**: LoRA r=16, alpha=32, 3 epochs, A10G 24GB
- **Mistral Nemo 12B**: Same config, L40S 48GB

Example: mayor says *'I will invest 50M in solar by round 3, but cut the environmental budget to fund it.'*

```
{  "promises": [    {"text": "invest 50M in solar", "type": "ecology
```

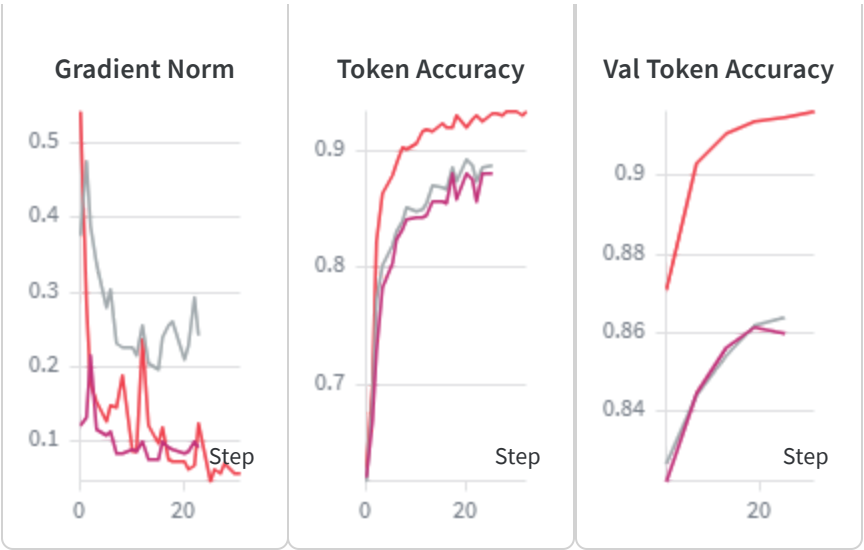
FT-Citizens

Mistral Small 24B: QLoRA 4-bit, r=16, alpha=32, 3 epochs, L40S 48GB. 351 examples.

```
{  "reactions": [    {"name": "Mia Chen", "type": "environmentalist"
```

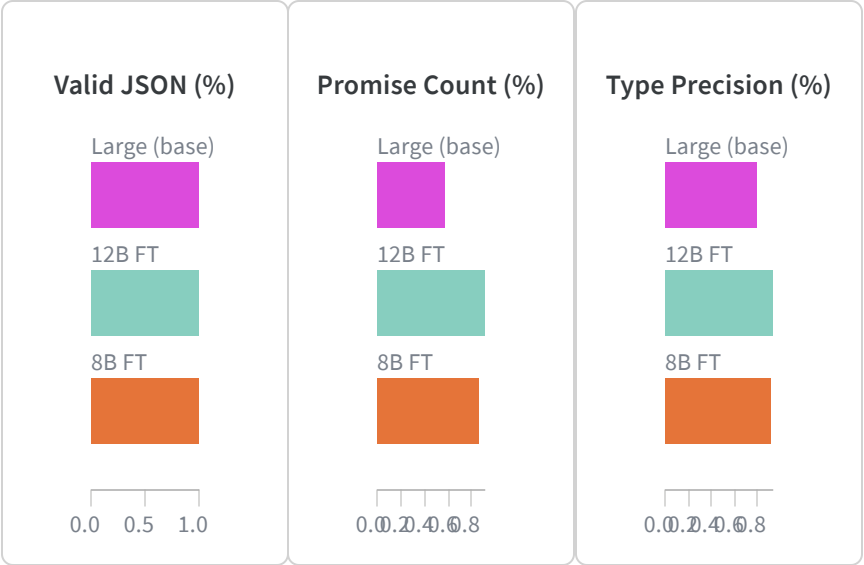
4. Training Results

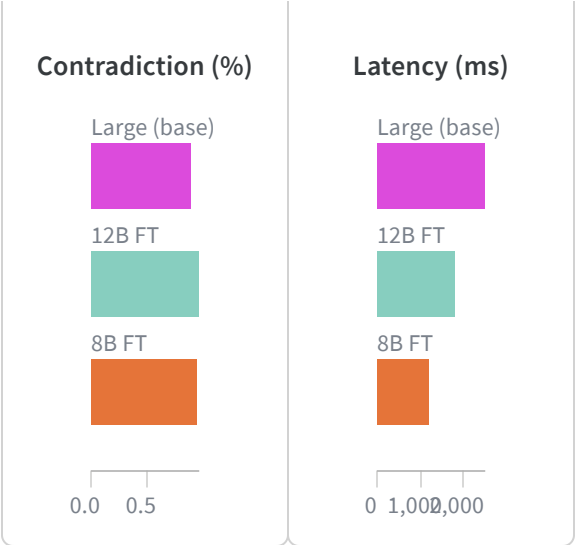
All three models converge within 3 epochs:



5. Evaluation: Promise Extraction

45 test examples (15 easy + 15 medium + 15 hard):

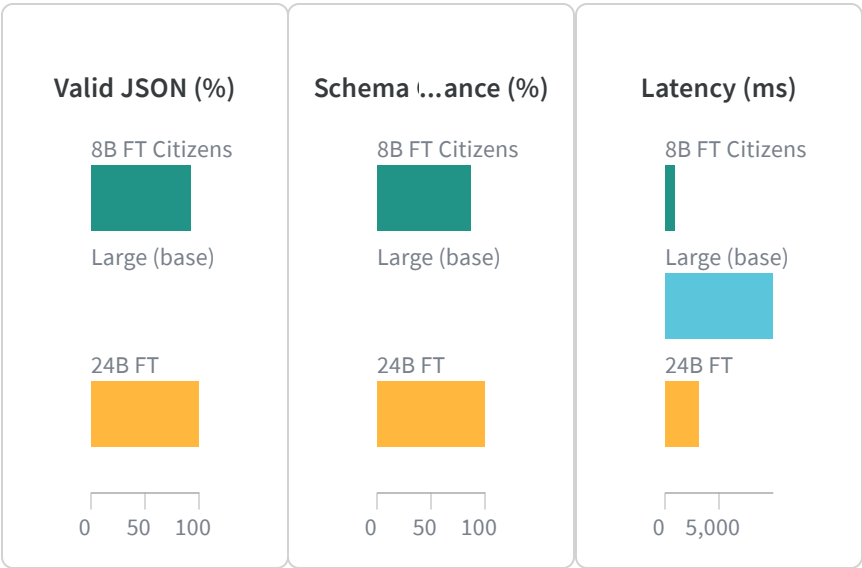




Both fine-tuned models beat Large on **promise count (+30%)** and **contradiction detection (+5%)**, at half the latency.

6. Evaluation: Citizen Reactions

39 validation examples. Strict JSON schema required for game engine:



Large returns markdown-wrapped JSON with wrong field names (**0% compliance**). The 8B FT gets 93%+ and 24B FT gets **100%**.

7. Key Findings

- Small fine-tuned models beat Large on both tasks for structured output
- 8B models are production-viable: 93%+ valid JSON, sub-second latency
- 24B FT: **100% schema compliance** vs 0% for Large on citizen reactions
- 8B/12B FT: **+30% promise count**, +5% contradiction detection vs Large
- QLoRA fits 24B on a single 48GB GPU, trains in under 10 minutes
- 690 Bedrock-generated examples are enough for strong task specialization

8. Resources

- Models: mistral-hackaton-2026/ecotopia-extract-ministral-8b (30.7MB LoRA)
- Models: mistral-hackaton-2026/ecotopia-extract-nemo-12b (39.4MB LoRA)
- Models: mistral-hackaton-2026/ecotopia-citizens-small-22b (69.8MB LoRA)
- GitHub: KyleKreuter/ecotopia

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Created with ❤️ on Weights & Biases.

https://wandb.ai/nolancacheux/hackathon-london-nolan-2026/reports/Ecotopia-Fine-Tuning-Mistral-for-Political-Simulation--VmlldzoxNjA2NzA3OA